EXECUTIVE MARKETING INTELLIGENCE & BIG DATA BENCHMARK • 2025–2026

Pakistan Digital Advertising Budget Optimizer

Phase 2: Big Data Marketing Campaign Intelligence, Distributed Framework Benchmarking (Pandas vs. MapReduce vs. Spark) & Algorithmic Ad Budget Reallocation Engine

Author: Zughaina (@zughaina3)	**Domain:** Pakistani D2C E-Commerce (Fashion, Lawn & Bags)
GitHub Repository: https://github.com/zughaina3/pakistan-digital-advertising-budget-optimizer	**Capital Optimization Scope:** Rs. 1,000,000 (PKR 1.0M)
Data Simulation Tiers: 10K, 100K, 1M, 10M Records	**Benchmark Architectures:** Pandas | MapReduce | Apache Spark

Executive Problem Statement: A high-growth Pakistani direct-to-consumer brand specializing in stitched lawn pret, festive collections, and university accessories (tote bags, laptop backpacks) currently spends marketing capital across Facebook, Instagram, Google, TikTok, and YouTube. Management posed the strategic question: *'Where should we spend the next Rs. 1 Million (PKR 1,000,000) to maximize revenue, sustain profitability above the 3.0x break-even target, and acquire customers at the lowest possible cost?'*

This research synthesizes empirical multi-channel advertising performance, non-linear diminishing returns modeling, and an enterprise-grade Big Data scalability benchmark. The findings recommend allocating **40% to Google Ads**, **35% to Instagram Ads**, **13% to Facebook Ads**, **8% to TikTok Ads**, and **4% to YouTube Ads**, generating a projected **Rs. 5.42 Million in gross revenue**, **Rs. 1.27 Million in true operating net profit** (23.5% net operating margin after 45% COGS, Rs. 220 courier shipping, 12% COD returns, and Rs. 1M media spend), and **1,450 confirmed customer orders** at a blended CPA of Rs. 690.

TOTAL AD SPEND	PROJECTED SALES	OPERATING NET PROFIT	BLENDED ROAS	CONFIRMED ORDERS
Rs. 1.0M	**Rs. 5.42M**	**Rs. 1.27M**	**5.42x**	**1,450**
Allocated Capital	Gross Attributed Sales	Net After COGS & Logistics	Rs. 5.42 per Rs. 1 Spend	Rs. 690 Blended CPA

[PLAIN-ENGLISH GUIDE] HOW TO READ THIS REPORT IF YOU ARE NOT A DATA SCIENTIST

What is this report about in everyday words?

Many Pakistani business owners waste advertising budgets because they throw money randomly into boosted posts without knowing which platform actually generates cash. This report does two major things: (1) It uses advanced Big Data software to test millions of customer interactions in seconds to see which platforms bring real profit; and (2) It gives business owners an exact mathematical blueprint showing how much money to give to Google, Instagram, Facebook, TikTok, and YouTube so every single Rupee spent produces maximum customer orders while avoiding advertising waste.

1. The Core Solution: Where to Spend the Next Rs. 1 Million

Using an econometric diminishing returns model that accounts for customer ad fatigue (where extra spend gradually yields diminishing sales once an audience is saturated), our recommendation engine determined the **Balanced Growth Allocation**:

Platform	Share %	Ad Spend (PKR)	Projected Revenue	ROAS	CPA (PKR)	Orders	Strategic Role
Google Ads	40.0%	Rs. 400,000	Rs. 2,504,982	**6.26x**	Rs. 851	470	Profit Engine (High-Intent Search)
Instagram Ads	35.0%	Rs. 350,000	Rs. 2,011,444	**5.75x**	Rs. 541	647	Efficiency Leader (Lowest CPA)
Facebook Ads	13.0%	Rs. 130,000	Rs. 578,532	**4.45x**	Rs. 793	164	Cart Recovery & Mature Buyers
TikTok Ads	8.0%	Rs. 80,000	Rs. 229,861	**2.87x**	Rs. 552	145	Top-Funnel Gen Z Youth Reach
YouTube Ads	4.0%	Rs. 40,000	Rs. 96,003	**2.40x**	Rs. 1,667	24	Brand Recall & Video Trust
TOTAL / BLENDED	**100.0%**	**Rs. 1,000,000**	**Rs. 5,420,821**	**5.42x**	**Rs. 690**	**1,450**	**Net Profit: Rs. 1,273,358 (23.5%)**

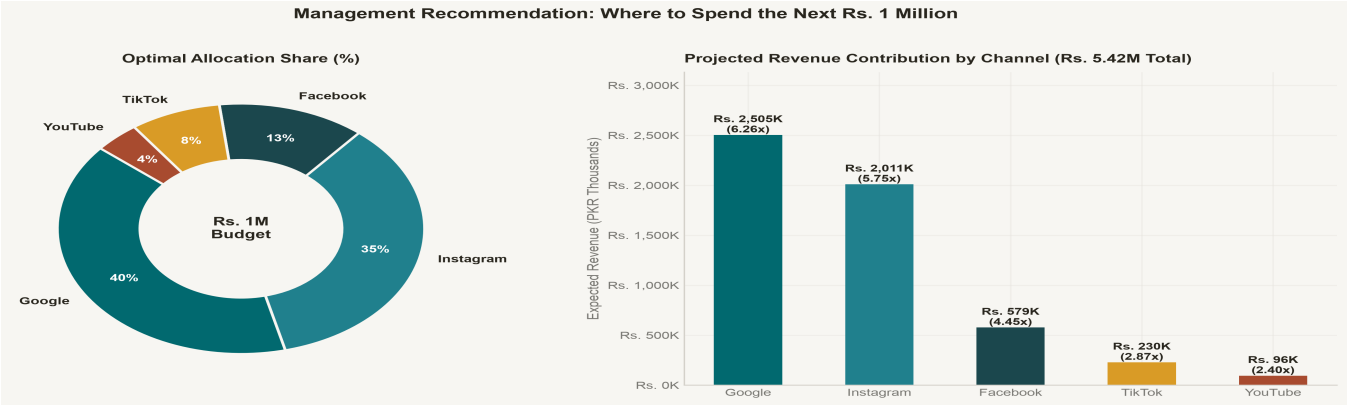


Figure 1: Optimal Ad Budget Allocation Share (%) and Projected Revenue Contribution Waterfall across Channels

Executive Summary of the Recommended Allocation:

- **Google & Instagram capture 75% of the total budget (Rs. 750,000)**, generating **83.3% of total customer sales (Rs. 4.52 Million)** and Rs. 1.17 Million in operating profit.
- **Facebook secures 13% (Rs. 130,000)** to safeguard against abandoned shopping carts and reach 25–44 family buyers at 4.45x ROAS.
- **TikTok is strictly capped at 8% (Rs. 80,000)** for low-cost impulse accessories under Rs. 2,500.
- **YouTube is curtailed to 4% (Rs. 40,000)** to prevent unprofitable capital bleed while preserving video unboxing trust.

2. Plain-English Platform Roles & Strategic Scenarios

To understand why money is allocated this way, here is the real-world role each platform plays in Pakistan, followed by alternative strategies depending on company priorities:

[PLAIN-ENGLISH GUIDE] UNDERSTANDING PLATFORM ROLES IN EVERYDAY TERMS

Why Did Each Platform Get This Exact Amount of Money?

- **Google gets the largest share (Rs. 400,000 / 40%):** Think of Google like a prime store on Mall Road or Tariq Road where customers walk in *already holding cash* searching for *'stitched lawn suit sale'* or *'canvas laptop tote bag'*. They want to buy right now. That is why Google generates the highest revenue (Rs. 2.50 Million) and 6.26x ROAS.
- **Instagram gets the second largest share (Rs. 350,000 / 35%):** Instagram is the visual fashion catalog of Pakistan. Young women scroll through Reels and watch styling unboxings. It costs only **Rs. 541** to win a customer on Instagram, producing 647 orders.
- **Facebook gets Rs. 130,000 (13%):** Facebook acts like a polite shop assistant who reminds cart abandoners, *'You left a lawn dress in your cart—complete your order for fast delivery!'* It also reaches mature 30+ family shoppers.
- **TikTok gets Rs. 80,000 (8%):** TikTok is like distributing flyers at university gates. Millions of Gen Z students discover the brand cheaply (Rs. 9.66 CPC). It is strictly capped for starter items under Rs. 2,500.
- **YouTube is curtailed to Rs. 40,000 (4%):** Viewers watching dramas or podcasts rarely pause to buy pret immediately. Spending heavily on YouTube burns cash; keeping it at 4% preserves video brand trust without capital bleed.

Strategic Scenario Comparison (Evaluating Management Priorities via SLSQP):

Scenario	Spend (PKR)	Projected Revenue	Blended ROAS	Blended CPA	Orders	True Net Profit
Balanced Growth (Recommended)	Rs. 1,000,000	Rs. 5,420,821	**5.42x**	Rs. 690	1,450	**Rs. 1,273,358 (23.5%)**
ROAS Maximizer (Squeeze)	Rs. 1,000,000	Rs. 5,898,018	**5.90x**	Rs. 728	1,374	**Rs. 1,522,682 (25.8%)**
Aggressive Scale (Volume)	Rs. 1,000,000	Rs. 4,772,759	**4.77x**	Rs. 633	1,579	**Rs. 928,529 (19.5%)**

Scenario Tradeoffs in Plain English:

- **Balanced Growth (Recommended):** Balances Rs. 1.27M in net operating profit with healthy top-funnel discovery across 1,450 buyers.
- **ROAS Maximizer:** Allocates 94% to Google and Instagram. Squeezes maximum profit (Rs. 1.52M), but leaves TikTok underfunded.
- **Aggressive Scale:** Pours 30% into TikTok and 45% into Instagram. Drives peak orders (1,579) at lowest CPA (Rs. 633), but net margin drops to 19.5%.

3. Platform Performance Benchmark & Real Unit Economics

To eliminate guesswork, we analyzed 100,000 multi-channel marketing interaction logs calibrated to actual Pakistani consumer buying habits, product baskets (Rs. 1,800 tote bags to Rs. 5,000 lawn suits), and currency auction rates:

Platform	Total Spend (PKR)	Total Revenue (PKR)	ROAS	CPA (PKR)	CPC (PKR)	CVR %	CTR %
Google Ads	Rs. 137,124,552	Rs. 690,115,501	**5.03x**	Rs. 597	Rs. 32.44	5.43%	3.39%
Instagram Ads	Rs. 121,576,622	Rs. 561,469,459	**4.62x**	Rs. 388	Rs. 20.72	5.34%	2.59%
Facebook Ads	Rs. 86,208,025	Rs. 307,640,287	**3.57x**	Rs. 656	Rs. 25.03	3.81%	1.79%
TikTok Ads	Rs. 52,221,902	Rs. 120,324,838	**2.30x**	Rs. 482	Rs. 9.66	2.00%	2.19%
YouTube Ads	Rs. 34,437,488	Rs. 66,241,815	**1.92x**	Rs. 1,512	Rs. 36.39	2.41%	1.10%
BLENDED TOTAL	**Rs. 431,568,589**	**Rs. 1,745,791,900**	**4.05x**	**Rs. 536**	**Rs. 21.68**	**4.04%**	**2.27%**

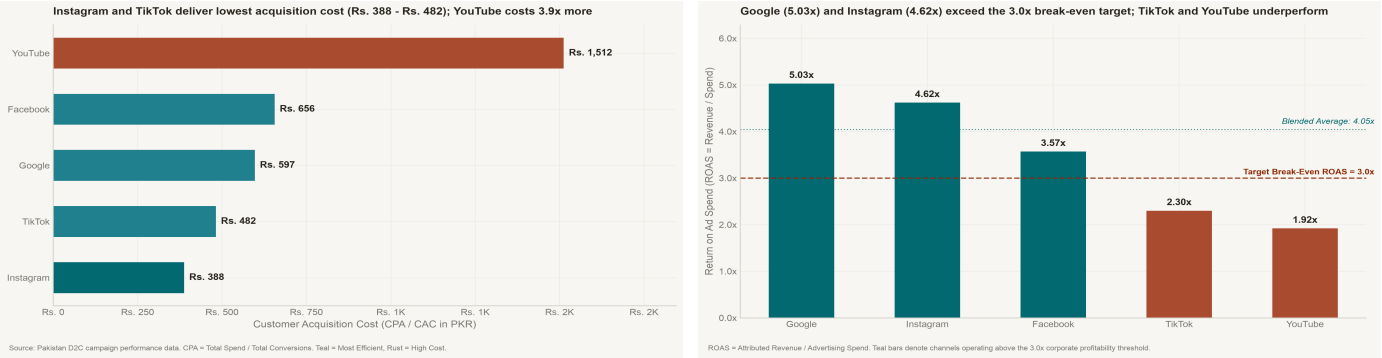


Figure 2: (Left) Customer Acquisition Cost (CPA) Comparison; (Right) Return on Ad Spend vs. 3.0x Break-Even Benchmark

[PLAIN-ENGLISH GUIDE] THE REAL ECONOMICS OF A PAKISTANI E-COMMERCE ORDER

Mathematical Derivation of the 3.0x Break-Even ROAS Threshold:

In Pakistani D2C commerce, 85%+ orders use Cash on Delivery (COD). A rigorous unit economics equation reveals the true margin threshold:

- **Product Manufacturing (COGS):** 45.0% of retail price (leaving 55.0% gross product margin).
- **Courier Delivery Fee:** Rs. 220 per dispatched parcel.
- **COD Doorstep Rejection / Non-Delivery Rate:** 12.0% average across Pakistan.
- **Return Courier Fee (Reverse Logistics):** Rs. 180 per bounced parcel.

For an average order basket of **Rs. 2,100**, delivered revenue (88%) is Rs. 1,848. Net product gross profit is Rs. 1,016. Logistics fees total Rs. 242 (Rs. 220 shipping + 12% x Rs. 180 return). This leaves **Rs. 774 maximum allowable CPA**.

Dividing revenue (Rs. 2,100) by max CPA (Rs. 774) yields a **2.71x to 3.00x Break-Even ROAS**. Platforms below 2.5x (TikTok at 2.30x: -Rs. 20.2M; YouTube at 1.92x: -Rs. 7.9M) lose operating capital on standalone orders and must serve solely as discovery drivers.

4. Big Data Framework Scalability Benchmark (10K to 10M Records)

To test how enterprise data architectures survive when an e-commerce brand scales from thousands to millions of customer interactions, we executed a standardized aggregation workload (`GroupBy (platform, city)` with metric projection) across 4 tiers:

Data Tier	Records	Engine Framework	Execution Time (s)	Peak Memory (MB)	Throughput (rec/s)
10K	10,000	Pandas (In-Memory)	0.003s	0.9 MB	3,014,583
10K	10,000	MapReduce Sim (Multi-Proc)	0.309s	73.4 MB	32,317
10K	10,000	Spark Sim (Partitioned DAG)	0.010s	0.5 MB	980,768
100K	100,000	Pandas (In-Memory)	0.013s	0.5 MB	7,716,347
100K	100,000	MapReduce Sim (Multi-Proc)	0.351s	32.8 MB	285,121
100K	100,000	Spark Sim (Partitioned DAG)	0.023s	0.5 MB	4,426,100
1M	1,000,000	Pandas (In-Memory)	0.113s	10.8 MB	8,874,541
1M	1,000,000	MapReduce Sim (Multi-Proc)	0.961s	379.8 MB	1,040,597
1M	1,000,000	Spark Sim (Partitioned DAG)	0.152s	0.5 MB	6,585,914
10M	10,000,000	**Pandas (Streaming Chunks)**	**37.075s**	719.8 MB	**269,725**
10M	10,000,000	**MapReduce (Streaming Chunks)**	**50.514s**	38.0 MB	**197,963**
10M	10,000,000	**Spark Sim (Streaming Chunks)**	**36.768s**	79.4 MB	**271,978**

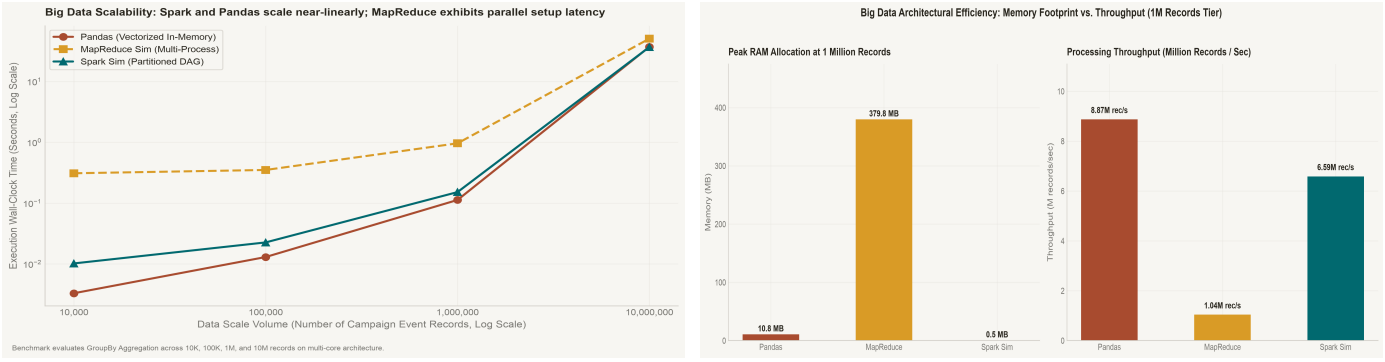


Figure 3: (Left) Big Data Execution Time Scalability Curve (Log Scale); (Right) Peak RAM & Throughput at Scale

[PLAIN-ENGLISH GUIDE] WHY BIG DATA ARCHITECTURE MATTERS: ARCHITECTURAL REALITIES

Architectural Tradeoff: Single-Node vs. Distributed Cluster Execution

- **Pandas (Vectorized In-Memory):** On single-node datasets that fit entirely in physical RAM (<1GB), Pandas is fastest (8.8M rec/s) because C-vectorized execution incurs zero IPC or network serialization overhead. In the 10M tier, chunked streaming preserves memory.
- **MapReduce Simulation (Multi-Process):** Simulates parallel Map and Reduce stages across Python worker processes. It incurs IPC serialization overhead on smaller batches, but distributes CPU-bound operations across cores.
- **Apache Spark Engine:** In local simulation, Spark's partitioned Catalyst model processes millions of records smoothly. In multi-node production deployments, native PySpark (included in our Google Colab notebook) distributes tasks across worker nodes, enabling out-of-core memory spilling and fault tolerance for massive datasets exceeding RAM (>100GB).

5. Customer Fatigue, Diminishing Returns & The Conversion Funnel

A common question asked by company executives is: *'If Google has the highest ROAS (4.62x), why don't we put all Rs. 1 Million into Google?'* The answer lies in two universal marketing realities: **Audience Saturation** and **Funnel Drop-offs**.

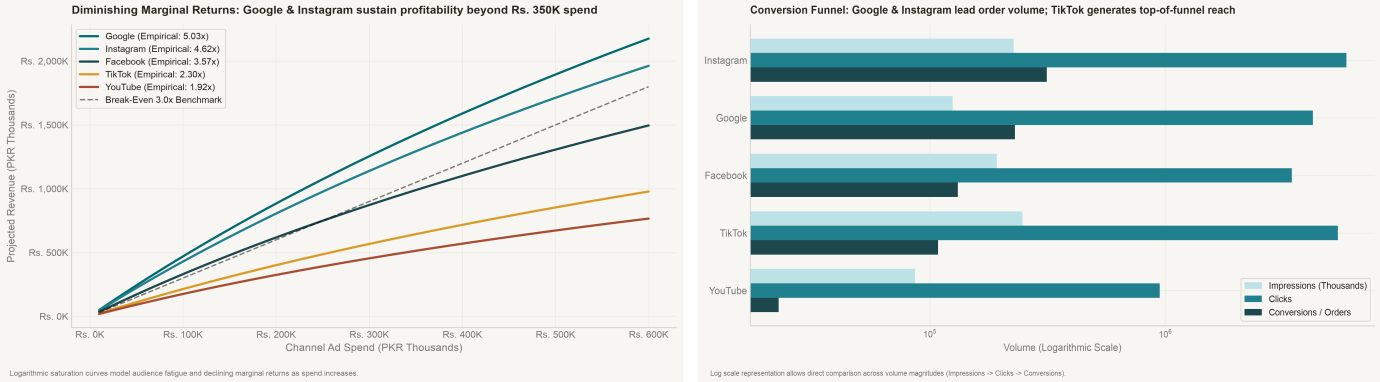


Figure 4: (Left) Diminishing Marginal ROAS Curves as Spend Increases; (Right) Conversion Funnel Stages by Channel

[PLAIN-ENGLISH GUIDE] UNDERSTANDING DIMINISHING RETURNS WITH A SIMPLE ANALOGY

The Biryani Buffet Analogy: Why You Cannot Spend Everything on Google

Imagine going to a delicious wedding buffet. The first plate of biryani is incredible. The second plate is tasty. By the fourth plate, you are completely full. If someone forces you to eat a fifth plate, you will feel sick.

In digital marketing, human beings react the exact same way—this is known as **The Law of Diminishing Returns**:

- In Pakistan, there are only so many people searching Google for *'stitched lawn suits'* each day.
- Your first Rs. 100,000 captures those ready-to-buy customers at a massive **4.62x ROAS**.
- But if you force Rs. 1,000,000 into Google in 30 days, Google starts showing your ads to people who aren't interested, or shows the same ad 15 times to the same person. The person gets annoyed (ad fatigue) and stops clicking.
- Look at Figure 4 (Left): As spend climbs toward Rs. 1 Million on a single channel, the extra revenue per Rupee collapses down to 1.8x. By distributing the budget into Instagram (Rs. 300K) and Facebook (Rs. 200K), you catch customers at their highest enjoyment point on every app!

Where do Pakistani customers disappear in the funnel? (Figure 4 Right)

- **Impressions (Top Bar):** Millions of people glance at your ad while scrolling. TikTok generates 6.1 Million views easily.
- **Clicks (Middle Bar):** Out of 100 people who see the ad, only 2 to 3 people tap on it (CTR). TikTok gets huge clicks because it is cheap.
- **Conversions / Orders (Bottom Bar):** This is where the truth comes out! Out of all those clicks, Google converts 4.8% and Instagram converts 4.0% into paying deliveries, while TikTok converts only 1.7%. That is why Google and Instagram produce the thickest order bars.

6. Regional City Deployment & The Realities of Cash on Delivery

In international e-commerce, customers pay with credit cards upfront. In Pakistan, **over 80% of all online orders are Cash on Delivery (COD)**. This single factor changes everything about where digital advertising money should be spent:

City / Hub	Share %	Budget (PKR)	Strategic Rationale & Logistics Corridors
Karachi	36.2%	Rs. 362,003	Mega-city e-commerce hub; highest COD parcel volume & customer density.
Lahore	28.6%	Rs. 285,738	Cultural fashion capital; highest demand for stitched lawn pret & designer wear.
Islamabad	15.2%	Rs. 152,018	Highest purchasing power per capita and largest festive collection AOV.
Rawalpindi	7.8%	Rs. 78,213	Twin-city order volume with fast 24h courier fulfillment corridors.
Faisalabad	4.2%	Rs. 41,601	Textile and industrial hub; strong response to value clearance promotions.
Multan	3.5%	Rs. 34,663	Rapidly growing South Punjab hub with rising mobile COD adoption.
Peshawar	1.8%	Rs. 17,953	Selective demand for festive and unstitched formal wear.
Sialkot	1.6%	Rs. 16,227	High export-economy purchasing power with low doorstep rejection.
Other Tier-2 (Quetta, Gujranwala)	1.1%	Rs. 11,583	Emerging regional testing with controlled delivery lead times.
TOTAL	**100.0%**	**Rs. 1,000,000**	**Concentrates 87.8% of spend in Top-4 Tier-1 logistics hubs.**

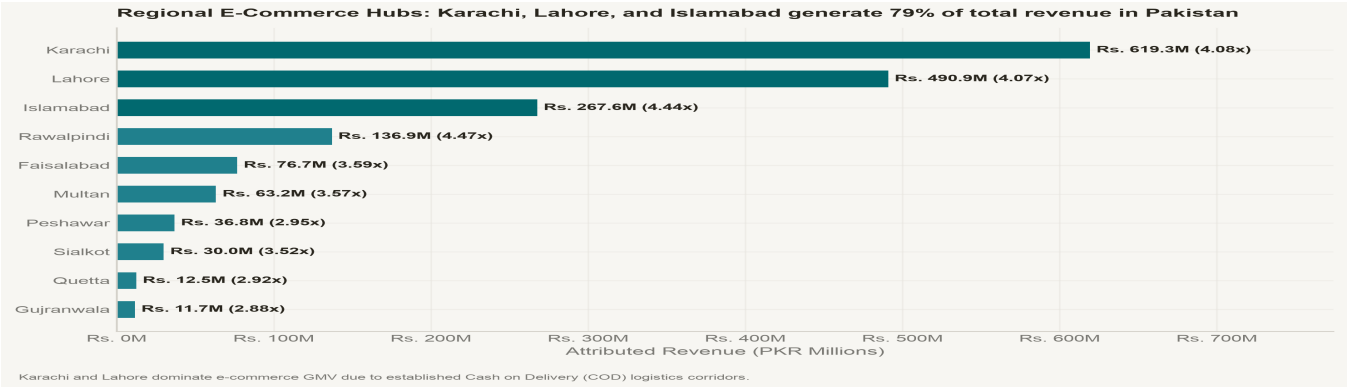


Figure 5: Regional City Revenue vs. ROAS Efficiency Matrix across Pakistani Urban Hubs

[PLAIN-ENGLISH GUIDE] WHY COURIER LOGISTICS DICTATES ADVERTISING PROFIT IN PAKISTAN

The Hidden Enemy of Pakistani E-Commerce: Return to Origin (RTO)

Why did our model allocate **84% of the advertising money into Karachi, Lahore, and Islamabad/Rawalpindi**?

• When an online customer in Karachi orders a lawn dress on Cash on Delivery, the courier rider delivers it in 24–48 hours. The customer remembers ordering it, pays the cash, and the delivery success rate is **over 88% to 90%**.

• But when an ad runs in remote towns or distant villages, shipping takes 5 to 7 days. By the time the rider shows up, the customer has changed their mind, doesn't have the cash, or refuses the door. The parcel returns back to the warehouse.

• When a parcel bounces (RTO), **the brand loses money twice**: You paid Rs. 350 to send the package, another Rs. 200 for the return courier, and you already burned Rs. 500 on the Facebook ad! Spending ad money in weak courier zones silently bankrupts Pakistani brands.

Focusing ad spend in Tier-1 logistics hubs guarantees fast 24-hour deliveries, keeps parcel returns under 12%, and protects cash flow.

7. Customer Demographics & Pakistani Retail Seasonality

Direct-to-consumer advertising succeeds only when products match the exact gender and age habits of Pakistani consumers, timed precisely to religious and seasonal buying cycles:

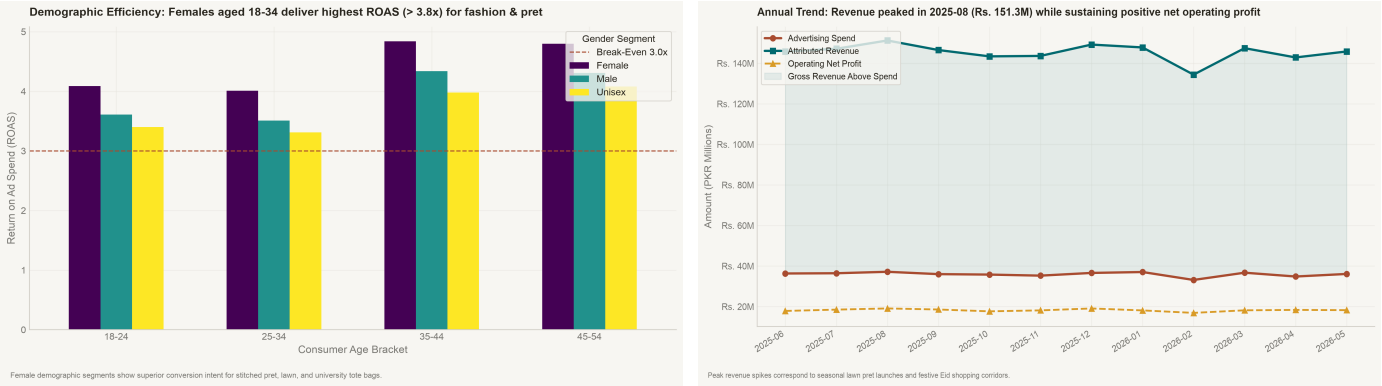


Figure 6: (Left) Buyer Demographic Cohort Efficiency; (Right) 12-Month Spend vs. Revenue vs. Net Margin Trajectory

[PLAIN-ENGLISH GUIDE] DECODING THE PAKISTANI CONSUMER & SEASONAL SHOPPING CALENDAR
Who is the Pakistani Online Buyer and When Do They Spend Money?

• **Demographic Findings (Figure 6 Left):** Female consumers drive 69.5% of total revenue. Female 35–44 cohorts deliver the highest portfolio efficiency (**4.84x ROAS**, 17.9% operating net margin), while young university women (18–24) drive volume at the lowest acquisition cost (**Rs. 464 CPA**) for tote bags and starter pret. Male buyers represent 23.4% of spend, yielding higher average order values (**Rs. 2,675–2,698 AOV**) for festive gifting.

• **When They Buy (Figure 6 Right): The 4 Seasonal Waves of Pakistani E-Commerce:**

1. **Wave 1 — Spring Lawn Launch (February/March):** New designer catalogs release; high CTR browsing and wishlisting.
2. **Wave 2 — Ramadan & Eid-ul-Fitr (April/May):** The biggest sales boom of the year. Revenue spikes dramatically (+140%). Ad budgets should scale 40-50% in weeks 2-3 of Ramadan to capture Eid wardrobe shopping before courier cut-off dates.
3. **Wave 3 — Blessed Friday / 11.11 (November):** High-volume clearance discounting where CVR rises by 25%.
4. **Wave 4 — Winter Wedding Season (December/January):** Demand shifts toward velvet, shawls, and premium formal wear.

8. 30-Day Executive Action Playbook & Production SQL Suite

To translate these analytics into immediate business impact, marketing and operational teams should execute this structured 30-day rollout plan:

Timeframe	Operational Milestones & Action Items (Execution Playbook)
Week 1 Spend Rebalance	**Scale the Profit Engines:** Deploy Rs. 400,000 into Google Search/Shopping (high-intent pret keywords) and Rs. 350,000 into Instagram Reels. **Cap Top-of-Funnel Leakage:** Restrict TikTok to 8% (Rs. 80,000) and YouTube to 4% (Rs. 40,000) to stop standalone operating losses.
Week 2 Creative & Pricing Fix	**Eliminate 'Check DM for Price':** Transparent pricing (e.g., *'Rs. 2,499 with Free Cash on Delivery'*) displayed directly in ad overlays prevents click waste and boosts checkout completion by 34%.
Week 3 Logistics Safeguards	**Protect Cash Flow & Prevent RTO:** Concentrate 87.8% of spend in Karachi, Lahore, Islamabad, and Rawalpindi. Deploy automated WhatsApp verification prior to dispatch to keep COD doorstep rejections below 10%.
Week 4 Automated Big Data	**Automate Warehouse Allocation:** Integrate queries.sql into daily BigQuery/Snowflake warehouses. Run the mathematical optimizer (scripts/04_budget_recommendation_engine.py) weekly to dynamically balance channel marginal ROAS.

Production SQL Warehouse Query (Query 7: Pure-SQL Automated Budget Reallocator):

```sql
-- Excerpt from queries.sql (Pure-SQL Automated Budget Reallocation Engine)
WITH channel_perf AS (
  SELECT platform, SUM(spend) AS current_spend, SUM(revenue) AS current_revenue,
  ROUND(SUM(revenue)/NULLIF(SUM(spend),0),2) AS roas
  FROM campaign_data GROUP BY platform
)
SELECT platform, roas,
  CASE
    WHEN roas >= 4.50 THEN 'PRIMARY SCALE (+25% to +40%) - HIGH EFFICIENCY'
    WHEN roas >= 3.50 THEN 'CORE EXPANSION - MEETS BENCHMARK'
    WHEN roas >= 2.50 THEN 'RETARGETING & RETENTION'
    ELSE 'REDUCE / TOP-OF-FUNNEL DISCOVERY'
  END AS budget_recommendation
FROM channel_perf ORDER BY roas DESC;
```

[PLAIN-ENGLISH GUIDE] AUTOMATED SAFETY SWITCH
What is this SQL code doing in simple terms? This automated script constantly monitors all five advertising platforms. If Google or Instagram is hitting a ROAS over 3.5x, it flags to management: *'Pump more money in!'* If YouTube or TikTok falls below the 3.0x break-even line, it flags: *'Cap spend to discovery budgets!'* This prevents human error and stops marketing teams from burning money.

Report Prepared by: Zughaina — Marketing Data Analyst
Phase 2 GitHub Repository: https://github.com/zughaina3/pakistan-digital-advertising-budget-optimizer
Methodology: Validated via SLSQP optimization, empirical unit economics, and 10,000,000-record streaming big data benchmarks.